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ORIGINAL ARTICLE

Mixture of orthogonal experts: a novel approach to multi-agent fast policy adaptation

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Abstract

Cooperative multi-agent reinforcement learning (MARL) has been widely applied in various complex decision-making domains due to its exceptional coordination capabilities. However, existing methods primarily focus on single-task scenarios or fixed-policy opponents (in competitive settings), making them less effective in non-stationary environments where tasks or opponent policies dynamically change. In this paper, we propose an algorithm termed MOFA, a novel method based on the centralized training and decentralized execution (CTDE) framework. By combining mixture of experts (MoE) and value function decomposition, it achieves fast policy adaptation in partially observable environments. Specifically, we integrate a shared-parameter MoE module into agent networks. Gram-Schmidt process is utilized to maintain the independence of expert subspaces, facilitating the extraction of transferable policy skills across diverse tasks. To enhance activation efficiency in expert modules, we use sparsemax to produce sparse probability distributions, ensuring only a few relevant experts are active at once. Since partial observability induces an information bottleneck, we maximize mutual information (MI) between local and global information as a solution. This is formalized through the optimization of a variational lower bound on the MI, which enhances decentralized agents' capability to infer global state features from limited local percepts. Experimental results demonstrate that, in two typical competitive environments, the MOFA algorithm exhibits significant advantages over multiple state-of-the-art algorithms in both multi-task learning and zero-shot generalization capabilities.

Keywords Multi-agent reinforcement learning · Non-stationary environments · Mixture of experts · Partially observable · Fast policy adaptation

Introduction

Multi-agent reinforcement learning (MARL) has seen remarkable progress in recent years. Its applications have expanded from early game strategy studies to domains such as multi-robot collaboration [1], video games [2], intelligent traffic management [3], and distributed energy systems [4]. Compared with single-agent reinforcement learning, however, MARL faces inherently more complex challenges. While existing research has made progress on issues such as partial observability [5], the curse of dimensionality [6], and credit assignment [7, 8], policy adaptation in non-stationary environments is becoming increasingly critical.

This challenge stems from the characteristics of real-world multi-agent systems. Practical applications often involve dynamic task objectives (e.g., fluctuating traffic patterns) or changing opponent behaviors in competitive settings. Such dynamic conditions expose the limitations of classi-

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cal MARL frameworks that assume stationary environments or fixed opponent policies.

Fast policy adaptation in these settings introduces two major difficulties. Agents must adjust to evolving task requirements or adversarial behaviors while maintaining coordinated actions. This process demands efficient perception of environmental changes, timely updates of decision strategies, and consistent coordination across agents. Classical MARL methods, however, often rely on inefficient sampling strategies and large amounts of interaction data, leading to high training costs and delayed adaptation.

Single-agent reinforcement learning has explored promising directions such as meta-learning [9, 10] and policy transfer or reuse [11–13]. In MARL, the non-stationarity problem is more complex due to interactions among multiple agents and dynamically changing opponents. It therefore requires addressing: 1) environmental non-stationarity, and 2) additional uncertainties introduced by variations in agent types, quantities, and behaviors.

Several recent MARL studies attempt to mitigate these challenges. UPDeT [14] employs a transformer-based dyn amic policy generator to handle varying entity inputs. REFIL [15] introduces entity-level value decomposition to filter irrelevant information, while OPT [16] leverages interaction prototypes to reduce noise and improve generalization. Beyond interaction modeling, system-level viewpoints have also been explored. Deng et al. [17] propose a neighborhood-coordinated actor-critic framework for adaptive traffic signal control, highlighting the difficulties of stable coordination in dynamic environments. Kavitha and Anusha [18] examine hierarchical clustering for decision modeling, showing that structural priors extracted through clustering may contrast with the assumptions of end-to-end MARL. These studies offer complementary perspectives but still face limitations in explicitly separating and reusing generalizable skills across tasks.

In this paper, we focus on the problem of fast policy adaptation in non-stationary multi-agent environments and propose MOFA, a mixture of orthogonal experts framework designed to extract and transfer generalizable skills across tasks. Built on the centralized training and decentralized execution (CTDE) value decomposition framework, MOFA departs from previous interaction-centric approaches by focusing on extracting and transferring generalizable skills across tasks. A parameter-shared mixture of experts (MoE) module is incorporated into each agent, and the Schmidt orthogonalization process is applied to enforce independence among expert subspaces, enabling clear separation of task-specific and task-general patterns. To improve expert activation efficiency, we employ sparsemax to generate sparse expert-selection distributions, reducing redundancy and improving computational efficiency. Furthermore, to mitigate the information bottleneck under partial observability, we maximize the mutual information between local observations and global states, enhancing the stability of decentralized execution.

Our contributions are summarized as follows:

- We introduce MOFA, the first MARL framework that integrates a parameter-shared MoE module with Schmidt orthogonalization into MARL for fast policy adaptation.
- We propose an orthogonal expert framework that extracts reusable skills across tasks, enabling effective decentralized execution without additional tuning.
- We demonstrate on StarCraft II and predator–prey benchmarks that MOFA achieves strong multi-task performance and robust zero-shot generalization.

The rest of paper is organized as follows: Section Related work presents the related work of CTDE-based MARL, fast policy adaptation and MoE, Section Preliminaries presents the problem formulation, and value decomposition algorithums. Section Method introduces the proposed MOFA algorithm, Section Experiments details validation environments, experimental settings, and corresponding experimental results, and finally Section Conclusion concludes the paper.

Related work

CTDE-based MARL

Centralized training with decentralized execution (CTDE) paradigm is a core framework in cooperative multi-agent reinforcement learning (MARL), enabling effective learning in decentralized partially observable Markov decision processes (Dec-POMDPs). Under CTDE, agents learn joint policies with access to global information during training but rely solely on local observations for execution. Existing CTDE methods can be broadly grouped into two classes. Value-based approaches decompose the joint value function into agent-wise utilities to support coordinated decision-making, with representative methods including VDN [7], QMIX [8], QPLEX [19], and COMA [20]. Policy gradient-based methods employ a centralized critic to estimate joint action-values and guide decentralized policy updates, as exemplified by MADDPG [21] and MAPPO [22]. Most of these methods assume a fixed task setting or stationary opponents. In contrast, this paper considers non-stationary environments where task configurations or opponent behaviors change over time, requiring rapid adaptation rather than learning a single policy. While our work builds upon the value decomposition framework, it departs from existing approaches by introducing mechanisms specifically designed

to support fast cross-task policy adaptation in dynamically varying scenarios.

Fast policy adaptation

Fast policy adaptation is a critical research direction in reinforcement learning. Its primary goal is to address efficient policy optimization for agents in dynamic environments. In single-agent, meta-reinforcement learning has emerged as a core method for achieving fast policy adaptation. Representative works such as MAML [9] optimize initial policy parameters, enabling agents to adapt to new tasks with minimal gradient updates. PEARL [10] further enhances cross-task transferability by incorporating latent variable modeling. Meanwhile, multi-task reinforcement learning also offers solutions to this challenge. For instance, the MOORE [23] algorithm innovatively employs a mixture of experts mechanism to explicitly decouple task features, enabling universal policy learning across multiple tasks through fixed parameters. MTDIFF [24], on the other hand, leverages diffusion models and transformer architectures, achieving cross-task policy planning and high-quality data synthesis via prompt learning and conditional reverse denoising processes without requiring additional parameter tuning. In multi-agent systems, fast policy adaptation faces dual challenges of environmental non-stationarity and partial observability. Existing methods primarily focus on decoupling entity relationships or separating input observations from policy distributions to achieve more generalizable value function representations [14–16]. Unlike conventional MARL approaches, our method leverages a mixture of experts mechanism to extract universal policy skills across tasks and dynamically adjusts the selection of expert networks based on local inputs, thereby enhancing model flexibility and adaptability. Notably, while inspired by MOORE [23], our approach differs by addressing MARL policy adaptation, sparsely selecting a small number of experts, and operating in more complex experimental scenarios. In addition, our method is trained based on the multi-task learning framework.

Mixture of experts

MoE represents a classical multimodal learning framework, initially introduced in [25] and [26]. Its core concept involves dynamically selecting or combining outputs from multiple expert networks through a gating network to handle complex tasks. MoE has achieved remarkable success in both natural language processing (NLP) and computer ision (CV). For instance, Shazeer et al.[27] introduced sparse MoE, which achieves efficient computation in large-scale language models. Chen et al.[28] integrated MoE into vision transformer, demonstrating superior performance in visual multi-task learning. Recently, MoE has been increasingly applied in reinforcement learning to address policy diversity and cross-task generalization challenges [23, 29]. For instance, DeepSeek-R1 [30] enhances reasoning capabilities through reinforcement learning and improves computational efficiency and reasoning performance by integrating the MoE architecture. In addition, MoE exhibits significant versatility, finding promising applications in recommendation systems [31], code generation [32], and mathematical reasoning [33], among other fields. Nevertheless, the application of the MoE architecture remains unexplored in MARL. This gap likely stems from the inherent complexity of MARL and the intricate interactions among agents, which complicate the adoption of MoE.

Preliminaries

Problem formulation

We model the cooperative MARL problem as a decentralized partially observable Markov decision process (Dec-POMDP), defined by a tuple $\langle \mathcal{E}, \mathcal{T}, \mathcal{N}, \mathcal{A}, \mathcal{S}, \mathcal{P}, \mathcal{R}, \Omega, O, \gamma \rangle$. Here, $\mathcal{E}$ represents the set of entities in the environment, $\mathcal{N} = \{1, 2, \ldots, n\}$ denotes the set of agents, and $\mathcal{A} = \{A^1 \times A^2 \times \cdots \times A^n\}$ represents the joint action space of all agents. Each entity e has a state representation s^e, and $s = \{s^e | e \in \mathcal{E}\} \in \mathcal{S}$ represents the global state of the environment. $\mathcal{T}$ is the set of tasks, where each task is composed of different number of entities and agents. At each time step t, each agent i receives a local observation $o_t^i \in \Omega$ according to the observation function $O(s_t, i)$ and selects an action $a_t^i \in A^i$, collectively forming a joint action $\boldsymbol{a}_t$. The environment then transitions to the next state s_{t+1} according to the transition function $P(s_{t+1}|s_t, \boldsymbol{a}_t)$. The joint policy π is determined by the joint action value function $Q(s_t, \boldsymbol{a}_t) = \mathbb{E}_{s_{t+1:\infty}, \boldsymbol{a}_{t+1:\infty}}[R_t | s_t, \boldsymbol{a}_t]$, where $R_t = \sum_{j=0}^{\infty} \gamma^j r_{t+j}$ is the discount return.

Value decomposition algorithms

We adopt value decomposition algorithms to learn policies. The key of the value decomposition method is to design an efficient mechanism. This mechanism is used to decompose the joint action-value function into individual action-value functions. A critical requirement for this decomposition is compliance with the Individual-Global-Max (IGM) principle [34]. This principle ensures that the optimal joint action can be determined by independently maximizing each agent's utility function. Formally, the IGM principle is represented as:

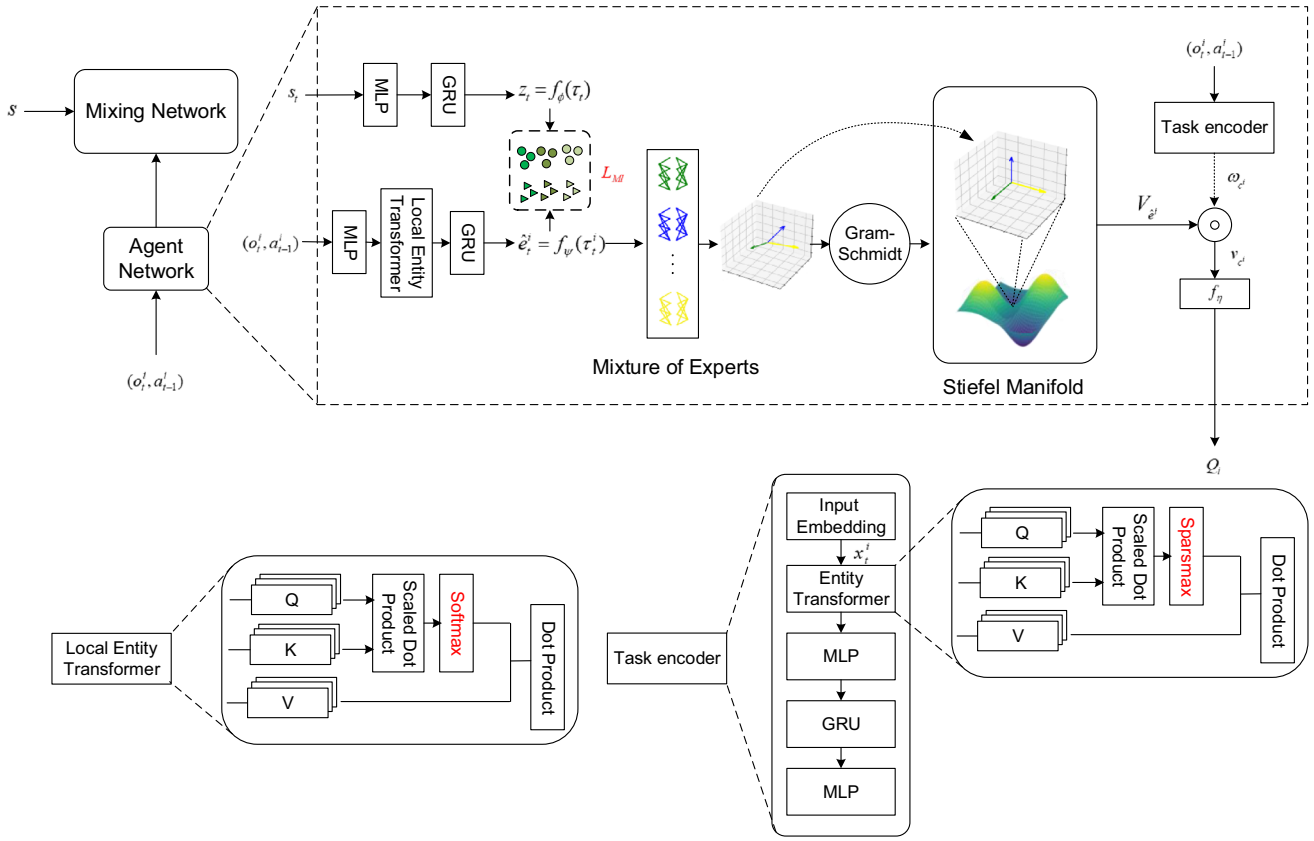


Fig. 1 The overall framework of MOFA

$$\arg\max_{\boldsymbol{a}_t} Q(o_t, \boldsymbol{a}_t) = \begin{pmatrix} \arg\max_{a_{1,t}} Q_{1,t}(o_{1,t}, a_{1,t}) \\ \arg\max_{a_{2,t}} Q_2(o_{2,t}, a_{2,t}) \\ \vdots \\ \arg\max_{a_{N,t}} Q_n(o_{N,t}, a_{N,t}) \end{pmatrix}, \quad (1)$$

where τ represents the joint action-observation histories of all agents, and $\boldsymbol{a}_t$ denotes the joint action. The IGM principle enables tractable computation of the greedy action selection by maximizing each agent's utility function locally.

To satisfy the IGM principle, QMIX [8] introduces a monotonicity condition:

$$\frac{\partial Q(o_t, \boldsymbol{a}_t)}{\partial Q_{i,t}(o_{i,t}, a_{i,t})} \geq 0, \forall i \in \mathcal{N}. \quad (2)$$

The QMIX employs a mixing network f to estimate the joint action value function as $Q(o_t, \boldsymbol{a}_t, s) = f(Q_{1,t}, Q_{2,t}, ..., Q_{1,N}, s)$, where the state s is incorporated as an additional input. The parameters of the mixing network are generated by hyper-networks that take the state s as input, and the network weights are constrained to remain non-negative to satisfy the monotonicity condition. This approach allows the utility functions of all agents to be updated efficiently using the temporal difference (TD) loss computed from the global action value function.

Method

Our research constructs a general policy with broad generalization capabilities via the multi-task learning framework. The core advantage of this policy is that when encountering new tasks similar to the training task distribution but not explicitly trained, the model achieves superior performance without fine-tuning. As shown in Fig. 1, our proposed MOFA algorithm framework is constructed on the CTDE paradigm. Under this framework, each agent learns an independent agent network by optimizing a mixed network's TD loss. Within each agent network (corresponding to each agent), we introduce an orthogonal representation extraction module based on the MoE mechanism, designed to improve the agent's ability to adapt quickly to task dynamic changes. Furthermore, to promote the agent's environmental awareness, we use global state information to guide the mixed expert representations' learning process through mutual information loss optimization.

Centralized situational awareness learning

The distributed policy typically relies on the local observation history of individual agents, which is limited compared to the holistic perspective provided by global information. This constraint leads to an incomplete representation of information in the orthogonal features extracted through the MoE model. Such limitations present significant challenges for developing universal strategies, which manifest in three main aspects: 1) Local observations fail to adequately capture the global environmental state; 2) Collaborative decision-making among agents often results in suboptimal policies; 3) The generalization capabilities of policies are substantially hindered.

To address these limitations, we introduce a parameterized global encoder f_ϕ and a shared-parameter local encoder f_ψ. By optimizing the mutual information (MI) loss [35, 36], we establish a progressive coupling between local information and global variations. Specifically, we define the global encoder as $z_t = f_\phi(\tau_t)$, where ϕ denotes the parameters of the global context encoder. We utilize a multi-layer perceptron (MLP) and a recurrent neural network (GRU) to encode the historical global state into τ_t, which is then passed into the global encoder f_ϕ. This encoder outputs a multivariate Gaussian distribution $\mathcal{N}(u_\phi(\tau_t^i), \sigma_\phi^2(\tau_t^i))$. By applying the reparameterization trick, we sample the global context vector $\hat{e}_t^i \sim f_\psi(\tau_t^i)$ from this Gaussian distribution. Similarly, the local encoder is defined as $\hat{e}_t^i = f_\psi(\tau_t^i)$, where ψ epresents the parameters of the local context encoder. Given the variability in entity cardinality and observation dimensionality across tasks, we adopt a hybrid architecture consisting of MLPs, transformer layers, and GRU units to encode historical observations and actions into τ_t^i. This vector is fed into the local encoder f_ψ, which outputs a multivariate Gaussian distribution $\mathcal{N}(u_\psi(\tau_t^i), \sigma_\psi^2(\tau_t^i))$. Again, we use the reparameterization trick to sample the local context vector $\hat{e}_t^i \sim f_\psi(\tau_t^i)$.

The MI $\mathcal{I}(\hat{e}_t^i; z_t|\tau_t^i)$ quantifies the information overlap between the local context $\hat{e}_t^i$ and the global context z_t given the local history τ_t^i, addressing the distributed policy's limitations such as the incomplete global perspective. However, direct computation of $\mathcal{I}(\hat{e}_t^i; z_t|\tau_t^i)$ is intractable due to the high-dimensional and complex nature of the joint distribution $p(\hat{e}_t^i, z_t|\tau_t^i)$. Thus, we apply variational inference to derive a tractable lower bound, following the information bottleneck framework. To approximate the intractable true posterior $p(\hat{e}_t^i, z_t|\tau_t^i)$, we introduce a parameterized variational estimator $q_\xi(\hat{e}_t^i|z_t, \tau_t^i)$, which is parameterized by ξ. Using the non-negativity of the Kullback–Leibler (KL) divergence ($D_{\mathrm{KL}}[q \parallel p] \geq 0$), we expand the KL divergence between q_ξ and the true posterior:

$$\mathbb{E}_{q_\xi}\left[\log \frac{q_\xi(\hat{e}_t^i|z_t, \tau_t^i)}{p(\hat{e}_t^i|z_t, \tau_t^i)}\right] \geq 0 \tag{3}$$

Rearranging this inequality yields:

$$\mathbb{E}_{q_\xi}\left[\log p(\hat{e}_t^i|z_t, \tau_t^i)\right] \geq \mathbb{E}_{q_\xi}\left[\log q_\xi(\hat{e}_t^i|z_t, \tau_t^i)\right]. \tag{4}$$

Substituting this into the equivalent form of MI (expressed as the difference between conditional entropies $\mathcal{I} = \mathbb{H}(\hat{e}_t^i \mid \tau_t^i) - \mathbb{H}(\hat{e}_t^i \mid z_t, \tau_t^i)$, we obtain the following tractable lower bound for the MI:

$$\begin{aligned}\mathcal{I}(\hat{e}_t^i; z_t|\tau_t^i) \geq &- \mathbb{E}_{(z_t,\tau_t^i)}[C\mathcal{E}[p(\hat{e}_t^i|z_t, \tau_t^i)\|q_\xi(\hat{e}_t^i|z_t, \tau_t^i)]] \\ &+ \mathbb{E}_{(z_t,\tau_t^i)}[H(\hat{e}_t^i|\tau_t^i)],\end{aligned} \tag{5}$$

where $C\mathcal{E}$ denotes the cross-entropy operator. Maximizing this lower bound pushes the variational estimator q_ξ to approximate the true posterior $p(\hat{e}_t^i|z_t, \tau_t^i)$, ensuring that the local context $\hat{e}_t^i$ and global context z_t share sufficient information. This leads to the desired progressive coupling between local and global variations. Finally, the MI objective is expressed as a sum of KL divergence expectations over all agents, which is easier to compute:

$$\mathcal{L}_{\mathrm{MI}} = \sum_{i=1}^{n} \mathbb{E}_D[D_{KL}[p(\hat{e}_t^i \mid \tau_t^i)\|q_\xi(\hat{e}_t^i \mid z_t, \tau_t^i)]]. \tag{6}$$

Representation of mixture of experts orthogonality

Value decomposition methods, such as QMIX [8], employ agent networks to aggregate individual Q-values into global Q-values. However, the conventional QMIX's agent network structure is relatively simplistic, struggling to capture the multimodal environmental dynamics in complex tasks. Inspired by the MOORE [23] in single-agent multi-task reinforcement learning, we propose integrating orthogonal expert representations into the agent network. This is achieved through Stiefel manifold constraints on expert representations, enhancing the model's expressive power and generalization capabilities.

Specifically, we feed the embedding vector $\hat{e}$, generated by the local encoder f_ψ into a MoE network comprising k experts. Each expert produces independent representations. Subsequently, we orthogonalize the expert outputs through the Gram-Schmidt process. Formally, we define the orthonormal representation as a matrix $V_{\hat{e}^i} = [\nu_1, ..., \nu_k] \in \mathbb{R}^{d\times k}$. Orthonormal representations belong to a topological space known as the Stiefel manifold. This manifold is smooth and differentiable, making it widely applicable in machine learning [37].

Definition 1 (Stiefel Manifold): The Stiefel manifold $\mathcal{V}_k(\mathbb{R}^d)$ is defined as the collection of all k-dimensional orthonormal vectors in Euclidean space, where $k \leq d$, $\mathcal{V}_k(\mathbb{R}^d) = \left\{ V_{\hat{e}^i} \in \mathbb{R}^{d \times k} : V_{\hat{e}^i}^T V_{\hat{e}^i} = I_k \right\}$.

From this perspective, our objective can be interpreted as finding a set of orthogonal representations on the Stiefel manifold. These orthogonal representations should effectively extract common features of the approximate true state space embedded in local information. Subsequently, we define the utility value $Q_i(\tau_i, a_i, c) = f_\eta(\varphi(f_\psi(\tau_i)) \cdot \omega_c^i) = f_\eta(\varphi(\hat{e}^i) \cdot \omega_c^i)$, where $\omega_c^i \in \mathbb{R}^k$ represents task-specific weights that combine the k-orthogonal representations into task-relevant one. As shown in Fig. 1, the weight parameter ω_c^i is generated by the task encoder module. The input to the task encoder module is agent i's local observation (o_t^i, a_{t-1}^i). f_ψ is an fuction with parameters ψ, which is used to generate the Q_i value. φ is a mapping function that maps $\hat{e}^i$ to a k-orthonormal representations $U_{\hat{e}^i} \in \mathbb{R}^{d \times k}$ by a MoE $h_\varrho = [h_{\varrho_1}, ..., h_{\varrho_k}]$ with parameters $\varrho = [\varrho_1, ..., \varrho_k]$, while ensuring that the generated representations are orthogonal to enforce diversity. This objective can be rigorously formulated as a constrained optimization problem. Here, hard constraints explicitly ensure orthogonality:

$$\begin{aligned} &\max_{\Theta=\{\varrho,\psi\}} J(\Theta) \\ s.t. \quad &h_\varrho^T(\hat{e}^i) h_\varrho(\hat{e}^i) = I_k, \end{aligned} \tag{7}$$

where $I_k \in \mathbb{R}^{k \times k}$ is the identity matrix. To achieve expert diversity without solving the constrained problem in Eq. 7, we orthogonalize the representations $U_{\hat{e}^i}$ using the Gram-Schmidt process.

Definition 2 (Gram-Schmidt Process): Gram-Schmidt (GS) process is an orthogonalization technique applied to a set of linearly independent vectors. It maps the vectors to a set of k-orthonormal vectors $V = v_1, ..., v_k : v_i \in \mathbb{R}^d, \forall i \leq k$ defined as:

$$v_k = u_k - \sum_{i=1}^{k-1} \frac{< v_i, u_k >}{< v_i, v_i >} v_i, \tag{8}$$

where the i-th expert's representation vector u_i is orthogonally projected onto the complement space of the subspace formed by the all $(i-1)$ experts' representations. We employ the GS process to map the expert-generated representations $U_{\hat{e}^i} = h_\varrho(\hat{e}^i)$ into orthogonal representations $V_{\hat{e}^i} = GS(U_{\hat{e}^i})$. This procedure satisfies the hard constraint in Eq. 7.

At each time step t, the observation and action (o_t^i, a_{t-1}^i) are first encoded using an entity embedding function and output the embedding feature x_t^i. In the subsequent analysis, the embedding matrix X will replace x_t as the primary representation to simplify the description. Assuming each expert represents an attention map where the relationship weights between entities are computed by the query with the corresponding key. The input embeddings $\mathbf{X}$ are transformed into three components: Query $\boldsymbol{Q}$, Key $\boldsymbol{K}$, and Value $\boldsymbol{V}$:

$$\begin{bmatrix} \mathbf{Q} \\ \mathbf{K} \\ \mathbf{V} \end{bmatrix} = \mathbf{X} \begin{bmatrix} \mathbf{W}_Q \\ \mathbf{W}_K \\ \mathbf{W}_V \end{bmatrix}, \tag{9}$$

where $\iota = \{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$ are parameter matrices and each matrix $W \in R^{dx \times dx}$ is trainable. The relationship weights between entities can be computed by taking the dot product of the queries and keys. Subsequently, an activation function is applied to normalize these weights into a probability distribution.

The softmax function is conventionally used as the normalization activation function in traditional transformer architectures [38, 39]. This function transforms input vectors into probability distributions through exponential operations, characterized by assigning non-zero probabilities to all possible categories. We implemented this structure in local encoder. However, in the task encoder, where the output represents each expert's participation weight, this strict positive probability allocation mechanism poses a significant issue: every expert is compelled to participate in computations, even when some contribute minimally to the current task. This "full participation" mechanism not only increases computational overhead but may also degrade model performance, as irrelevant experts' participation can introduce noise interference, thereby affecting decision-making accuracy. Consequently, a sparsity mechanism is critical to improve model efficiency and performance. By dynamically deactivating specific experts according to task demands, it optimizes computational resource allocation and strengthens the model's generalization ability. Building upon the principles of sparse representation optimization outlined in [16] and [40], we adopt the sparsemax function to replace the softmax function. The key advantage of sparsemax is its ability to generate sparse probability distributions, a feature not present in softmax:

$$\text{sparsemax}(\overline{z}) := \arg\min_{\boldsymbol{p} \in \Delta^M} \|\boldsymbol{p} - \overline{z}\|_2, \tag{10}$$

where $\overline{z} \in \mathbb{R}^M$ is the input vector and $\boldsymbol{p} \in \mathbb{R}^M$ is the output vector. $\Delta^M := \{\boldsymbol{p} \in \mathbb{R}^M | \boldsymbol{p} \geq 0, \|\boldsymbol{p}\|_1 = 1\}$ is the probability simplex. The solution of Eq. 10 is obtained as follows:

$$\boldsymbol{p} = [\overline{z} - \sigma(\overline{z})1]_+, \tag{11}$$

where $[.]_+$ is a clipping function and $1 \in \mathbb{R}^M$ enotes the all-one vector. $\sigma(.) : \mathbb{R}^M \longrightarrow \mathbb{R}$ is a threshold function. the

threshold value is given by:

$$\sigma(\overline{z}) = \frac{(\sum_{i \le m(\hat{z})} \hat{z}_i - 1)}{m(\hat{z})}. \tag{12}$$

This allows us to directly compute the sparse probability distribution based on the threshold.

CTDE policy learning

We adhere to the CTDE paradigm, enabling agents to learn their individual utility functions by optimizing the joint action-value function for credit assignment. During the centralized training phase, the local action value of all agents is fed into a mixing network to estimate the global action value Q_{tot}. This training scheme simultaneously promotes situational awareness and improves MoE representation learning. Therefore, the total training loss of the MOFA model consists of two key components: temporal difference (TD) loss and MI loss. The TD loss optimizes the joint action-value function of the mixing network, guiding agents toward their optimal policies:

$$\mathcal{L}_{TD}(\theta) = \mathbb{E}_{\mathcal{D}}[(y_{tot} - Q_{tot}(s, \boldsymbol{a}; \theta_v))^2], \tag{13}$$

where $\mathcal{D}$ represents the experience replay buffer, $y^{tot} = r + \gamma \max_{u'} Q_{tot}(s', \boldsymbol{a}', \theta_v^-), \theta_v^-$ is the parameter of the target network. Meanwhile, the MI loss promotes agents' collaborative perception of environmental dynamics by maximizing the correlation between local observations and global states. Thus, given the two corresponding loss items, the total loss of our framework is formulated as follows:

$$\mathcal{L}_{tot} = \mathcal{L}_{TD} + \alpha_{MI}\mathcal{L}_{MI}, \tag{14}$$

where α_{MI} is coefficient. The overall framework is trained in an end-to-end centralized manner, which is summaried as Algorithm 1.

Experiments

We tested our method in two classic environments: StarCraft multi-agent challenge[1] [41] and predator–prey [16]. SMAC evaluates multi-task performance, while PP assesses zero-shot generalization. All experiments were implemented using PyMARL [41], a Python-based multi-agent reinforcement learning framework. The hyperparameters of the baselines match PyMARL's original implementation. For specific configurations, refer to the source code. All of the results in this paper are completed on NVIDIA 4090D GPUs.

[1] We use SC2.4.10 version.

Algorithm 1 The training process of MOFA

Require: Agent network θ_π, mixing network θ_v, target network $\theta_v^- = \theta_v$, global encoder f_ϕ, local encoder f_ψ, replay buffer $\mathcal{D}$

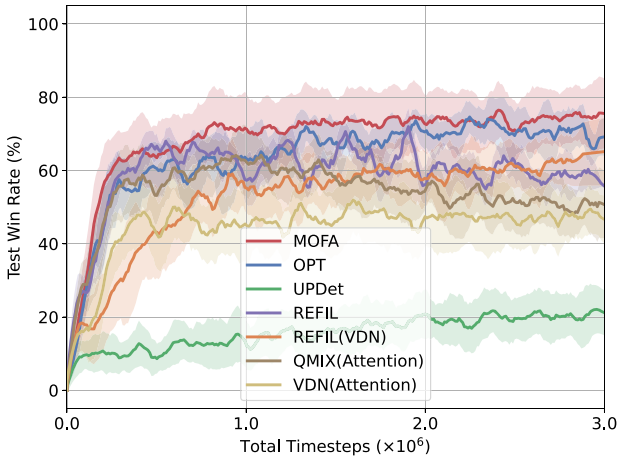
for $episode = 1, ..., M$ **do**
  Sample a task from $\mathcal{T}$
  for $t = 0, ..., T$ **do**
    Receive global state s_t
    Receive local observation $o_t^i, i = 1, ..., n$
    Encode the historical global state and the corresponding action-observation into τ_t and $\tau_t^i, i = 1, ..., n$
    $z_t = f_\phi(\tau_t), \hat{e}_t^i = f_\psi(\tau_t^i), i = 1, ..., n$
    $U_{\hat{e}^i} = h_\varphi(\hat{e}^i), i = 1, ..., n$
    $V_{\hat{e}^i} = GS(U_{\hat{e}^i}), i = 1, ..., n$
    $v_{c^i} = V_{\hat{e}^i} \cdot w_{c^i}, i = 1, ..., n$
    $Q_i = f_\psi(v_{c^i}), i = 1, ..., n$
    Sample joint action $\boldsymbol{a}_t$ from Q based on ϵ-greedy
    Receive the next joint observation o_{t+1}, next state s_{t+1}, and reward r_{t+1}
    Store $(o_t, a_t, r_{t+1}, o_{t+1}, s_t, s_{t+1})$ into replay buffer $\mathcal{D}$
  end for
  Randomly sample batch data from $\mathcal{D}$
  Calculate the joint action-value Q with mixing network, compute the update target y_{tot} and the TD loss in Eq. 13
  Calculate $\mathcal{L}_{tot}$ by Eq.14
  Update the whole network by minimizing the total loss $\mathcal{L}_{tot}$ with gradient descent
  Replace target networks with target update interval
end for

More detailed parameter information about the training process of MOFA is elaborated in Appendix A:.

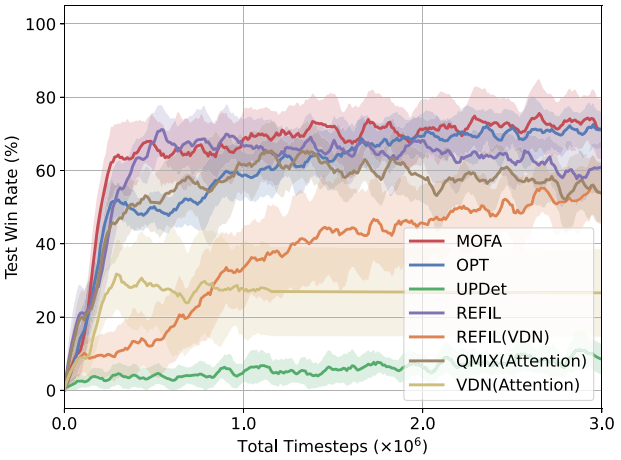
Environments description

StarCraft multi-agent challenge (SMAC): Our experiments cover six multi-task SMAC scenarios, including three foundational settings (3–8csz, 3–8MMM, 3–8sz) introduced by REFIL [15] and three extended, more complex settings (5–11csz, 5–11MMM, 5–11sz) proposed by OPT [16]. The multi-task configurations across the six scenarios are as follows:

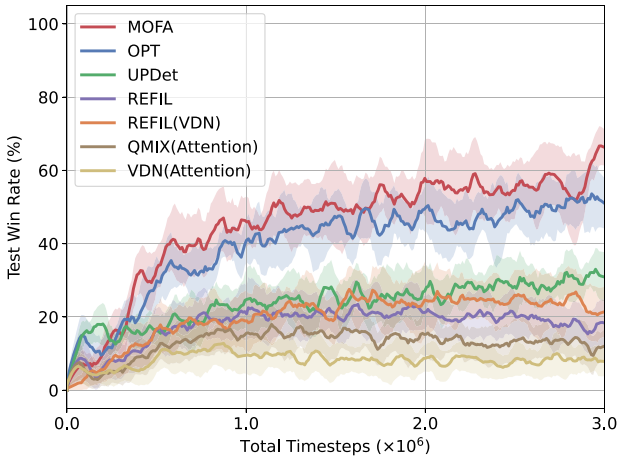
1) 3–8csz contains two symmetrical teams, each containing 3–6 Stalker/Zealot (S/Z) hybrids and 0–2 Colossi, resulting in 66 unique tasks;
2) 3–8sz contains two symmetrical teams, with each team consisting of 3–8 Stalkers/Zealots, resulting in 39 unique tasks;
3) 3–8MMM contains two symmetrical teams, with each team consisting of 3–8 Stalkers/Zealots and 0–2 Medics, resulting in 66 unique tasks;
4) 5–11csz contains two symmetrical teams, each with 5–8 Stalkers/Zealots and 0–3 Colossi, yielding 120 unique tasks;
5) 5–11sz contains two symmetrical teams, each comprising 5–11 Stalkers/Zealots, resulting in 63 unique tasks;



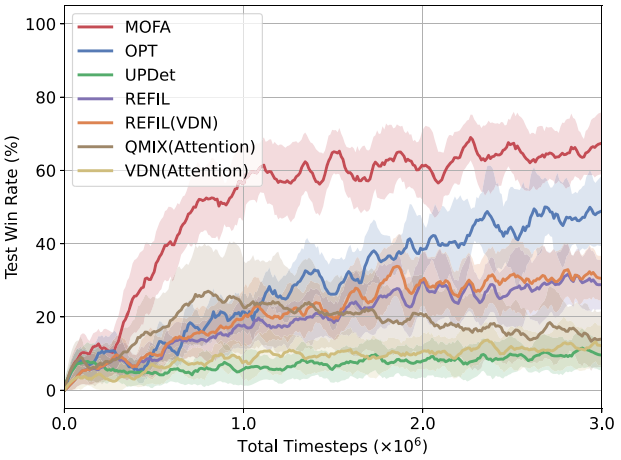
(a) 3-8csz (66 Tasks, Easy)



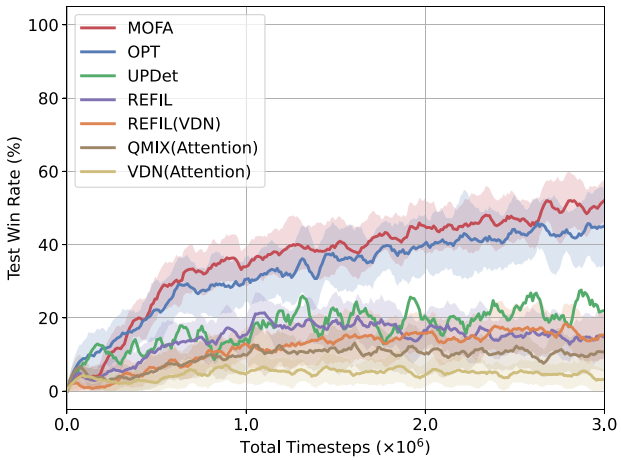
(b) 5-11csz (120 Tasks, Easy)



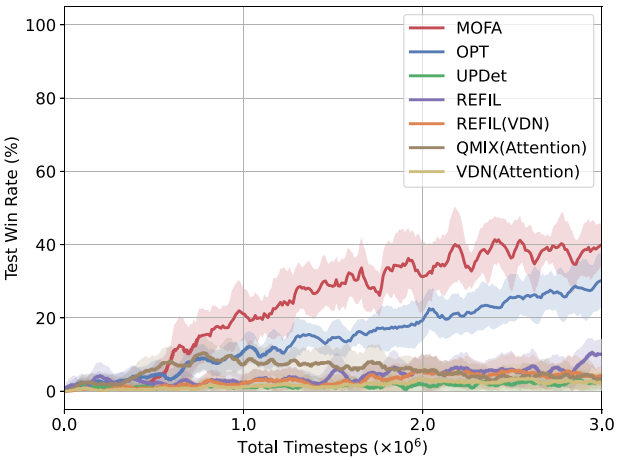
(c) 3-8MMM (66 Tasks, Hard)



(d) 3-8sz (39 Tasks, Hard)

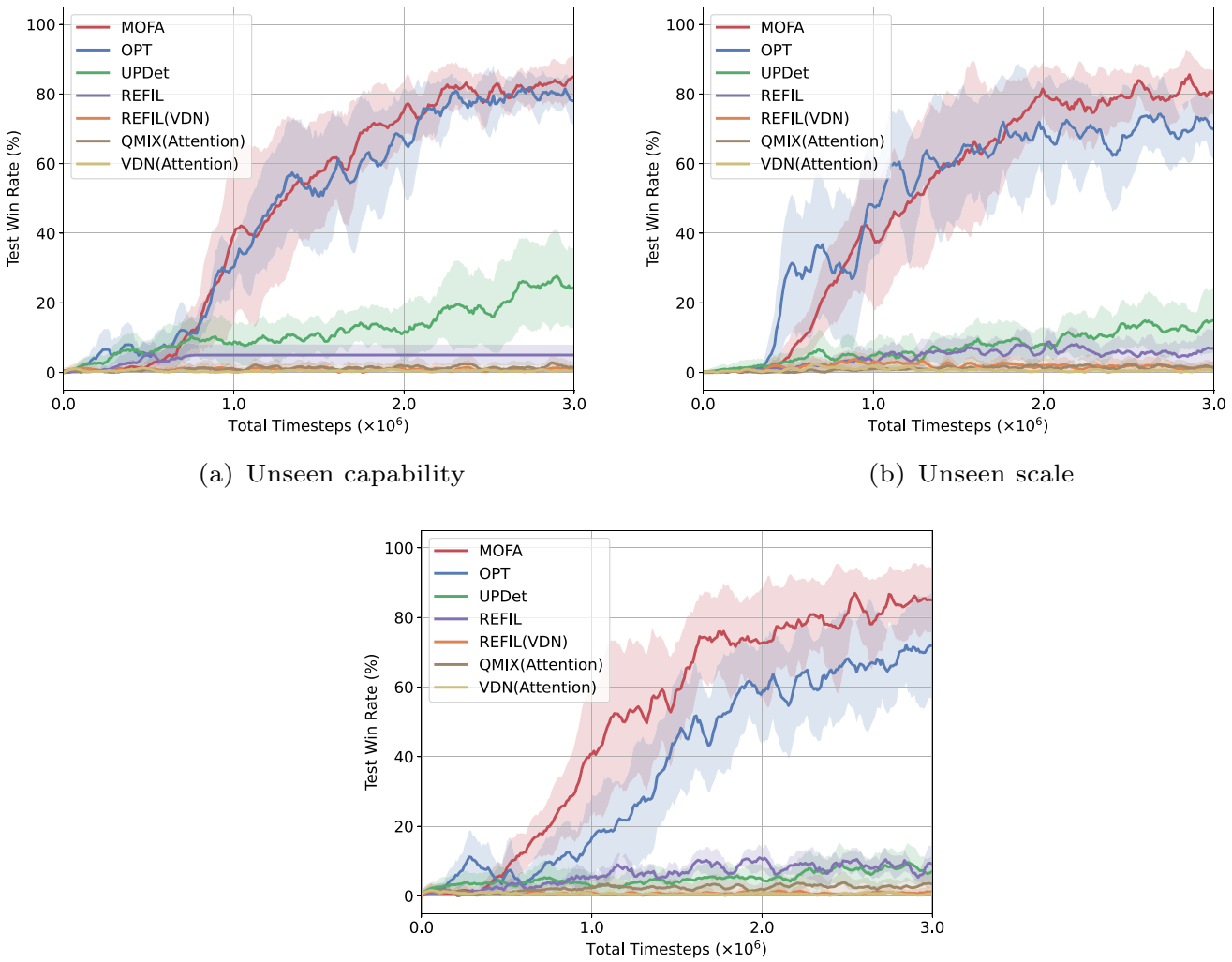


(e) 5-11MMM (120 Tasks, Super Hard)



(f) 5-11sz (63 Tasks, Super Hard)

Fig. 2 Learning curves of our method and baselines in 6 multi-task SMAC scenarios



(a) Unseen capability

(b) Unseen scale

(c) Unseen C & S

Fig. 3 Learning curves of our method and baselines in 3 multi-task PP scenarios, where the results are provided by testing the trained agents on the unseen tasks

6) 5–11MMM contains with two symmetrical teams, each consisting of 5–8 Marines/Marauders and 0–3 Medics, producing 120 unique tasks. These scenarios are classified into three difficulty levels: Easy (3–8csz, 5–11csz), Hard (3–8MMM, 3–8sz), and Super Hard (5–11MMM, 5–11sz).

Predator–prey (PP): We adopt the predator?prey environment introduced by OPT [16] to evaluate the zero-shot generalization performance of MOFA. Specifically, N_a predators, N_p prey, and no fixed obstacles are initialized in a grid world, where $\{N_a, N_p, N_o\}$ defines the scale of each task. Additionally, each predator is equipped with an attack capability C_a, and each prey possesses a defense capability C_p. At each time step, predators with limited sight range can move, stop, and capture, while prey move randomly and obstacles block movement. The goal of the predators is to capture all prey, where a prey can only be captured if the total attack capability of the predators is greater than or equal to the prey's defense capability. During training, the initial scale and capabilities are randomly sampled for each task. We conduct experiments on three multi-task PP scenarios, testing the agents in tasks with: 1) Unseen Capability (fix the scale parameters and randomly generate $\{C_a, C_p\}$), 2) Unseen Scale (fix the capability parameters and randomly generate $\{N_a, N_p, N_o\}$), and 3) Unseen Capability and Scale (C & S, randomly generate the scale and capability parameters).

Baselines

Our evaluation includes six baseline algorithms:

Table 1 The area under curve of MOFA and baselines in 6 multi-task SMAC scenarios (mean ± std). The best result is highlighted in boldface

Algorithms	3–8 csz	3–8 MMM	3–8 sz	5–11 csz	5–11 MMM	5–11 sz
UPDeT	0.16±0.01	0.24±0.01	0.07±0.01	0.06±0.0	0.18±0.01	0.02±0.00
OPT	0.64±0.02	0.39±0.02	0.3±0.05	0.6±0.02	0.33±0.02	0.15±0.02
REFIL	0.59±0.0	0.18±0.01	0.2±0.03	0.61±0.01	0.14±0.01	0.04±0.01
REFIL(VDN)	0.52±0.03	0.2±0.03	0.22±0.03	0.37±0.07	0.12±0.03	0.03±0.01
VDN(Attention)	0.44±0.03	0.08±0.02	0.09±0.01	0.33±0.08	0.05±0.01	0.02±0.01
QMIX(Attention)	0.54±0.01	0.13±0.01	0.18±0.08	0.55±0.01	0.09±0.02	0.06±0.01
MOFA	**0.69±0.02**	**0.45±0.04**	**0.52±0.03**	**0.66±0.02**	**0.37±0.03**	**0.25±0.03**

Table 2 The area under curve of MOFA and baselines 3 multi-task predator–prey scenarios (mean ± std), where the results are provided by testing the trained agents on the unseen tasks. The best result is highlighted in boldface

Algorithms	Unseen capability	Unseen scale	Unseen C & S
UPDeT	0.12±0.03	0.07±0.02	0.05±0.02
OPT	0.46±0.06	0.49±0.11	0.38±0.07
REFIL	0.06±0.03	0.04±0.00	0.05±0.02
REFIL(VDN)	0.01±0.00	0.02±0.00	0.01±0.00
VDN(Attention)	0.0±0.00	0.01±0.00	0.01±0.00
QMIX(Attention)	0.01±0.00	0.01±0.00	0.02±0.01
MOFA	**0.49±0.08**	**0.51±0.08**	**0.49±0.04**

UPDeT [14] is a transformer-based MARL method that removes the fixed input/output constraints inherent in RNN architectures for multi-agent tasks. Its attention-driven design enhances representation capacity and substantially improves cross-task generalization.

OPT [16] applies a sparse activation function and contrastive disagreement loss for extracting interaction prototypes. Then, it restructures these prototypes via an aggregator. By maximizing the mutual information between the aggregation weights and historical behaviors, OPT stabilizes the training process. These operations enhance the model's generalization ability and interpretability.

REFIL [15] is a novel entity decomposition approach that employs attention masking to process variable-sized entity inputs. This mechanism enables flexible adaptation to task scenarios with differing scales through dynamic input handling, eliminating conventional requirements for fixed entity quantities. The proposed architecture achieves cross-task adaptability with a unified framework, overcoming traditional methods' dependence on predetermined entity configurations.

REFIL(VDN) [15] modifies the Q-value synthesis method of the base version by replacing QMIX's nonlinear mixing network with VDN's linear summation structure.

QMIX(Attention) [8] is the QMIX algorithm with an introduced attention mechanism, which can handle a variable number of entity inputs.

VDN(Attention) [7] is the VDN algorithm with an introduced attention mechanism, which can handle a variable number of entity inputs.

Multi-task environments analysis

This section evaluates the performance of MOFA in multi-task scenarios using SMAC, with comparative analysis against baseline methods. Figure 2 displays the dynamic learning curves across six multi-task scenarios, while the corresponding Area Under the Curve (AUC) metrics are presented in Table 1. As shown in Fig. 2, MOFA exhibits noticeably faster learning progress during the early stages of training. Across all scenarios-ranging from easy (3–8 csz, 5–11 csz) to hard (3–8 MMM, 3–8 sz) and super-hard settings (5–11 MMM, 5–11 sz), MOFA reaches relatively high win rates within the first 0.5−1.0 million timesteps. This trend contrasts with the slower improvement of OPT, REFIL, UPDeT, and the attention-based value decomposition methods, suggesting that the expert-mixture architecture allows MOFA to adapt more efficiently during multi-task training. In addition to the faster initial learning, MOFA also achieves higher overall performance in most evaluated scenarios. As summarized in Table 1, MOFA attains the best AUC results in five out of the six settings, indicating that it maintains stronger performance throughout the full training process. As task difficulty increases, the performance differences become more apparent. In the super-hard scenarios (Fig. 2eand Fig. 2f), many baselines converge to low win rates and show limited progress, whereas MOFA continues to improve over time and ultimately reaches higher final win rates. This pattern reflects the stability of MOFA?s orthogonal-expert design under more challenging multi-task conditions. Furthermore, although prototype-based (OPT)

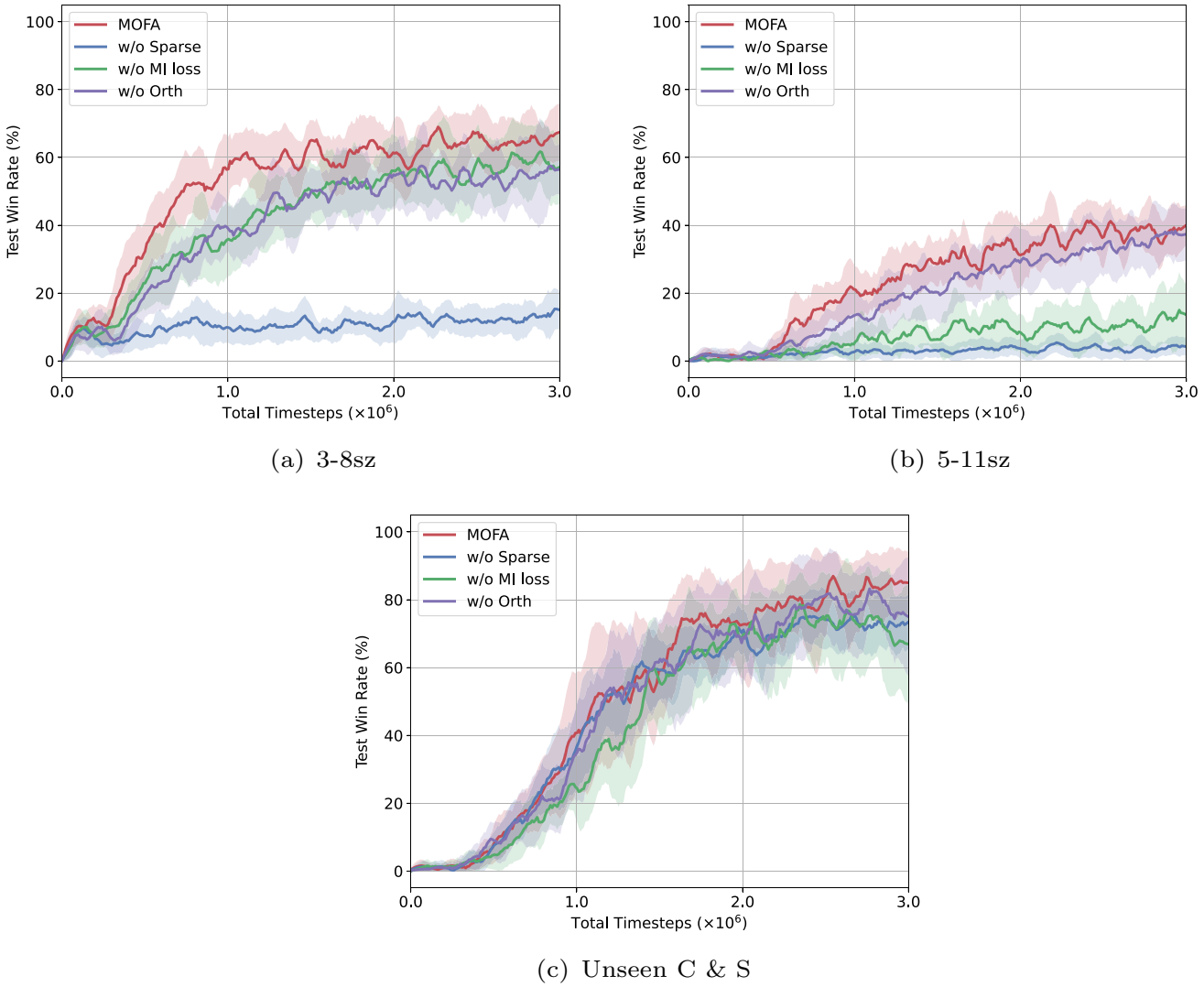


Fig. 4 Ablation study on three major components

and attention-based entity-processing approaches (REFIL, QMIX/VDN with attention) achieve moderate gains, they remain consistently below MOFA in both learning speed and final performance, indicating that representation enhancements alone may not be sufficient for effective multi-task policy learning. Overall, Fig. 2 and Table 1 show that MOFA achieves faster learning, more stable training behavior, and higher final performance across a diverse set of SMAC multi-task environments.

Zero-shot generalization analysis

We conducted a comparison of the performance of multiple baseline algorithms in the zero-shot generalization games of the PP setting, Fig. 3 shows the learning curves on unseen tasks, while Table 2 summarizes the corresponding AUC values. The results of the comparison indicate that there are significant differences in the generalization capabilities of various models when faced with unseen task settings. Across all three types of unseen tasks, unseen capability and unseen scale, and the combined setting, MOFA consistently reaches higher win rates throughout training, as reflected in the learning curves of Fig. 3. This trend is also supported by Table 2, where MOFA achieves the highest AUC values in all three categories. OPT also displays relatively strong generalization performance. Its learning curves rise more substantially than those of the remaining baselines, and its AUC values rank second across all three settings. These results indicate that OPT preserves part of the task-transfer ability learned during multi-task training. In contrast, the other baselines show limited improvement when evaluated on unseen tasks. As shown in Fig. 3, their win rates remain low across the full training horizon, and Table 2 further confirms that their AUC scores stay close to zero in most cases. This suggests

that these methods face difficulty adapting to task variations not encountered during training. Overall, Fig. 3 and Table 2 highlight that MOFA generalizes more effectively to unseen pp tasks, followed by OPT, whereas the remaining baselines show limited zero-shot transfer under the same evaluation protocol.

Ablations

We conduct ablation experiments to examine the contribution of three key components of MOFA: sparsemax activation, mutual information loss, and Schmidt orthogonalization. The evaluated variants include: 1) MOFA without sparsemax (w/o Sparse), 2) MOFA without mutual information loss (w/o MI loss), and 3) MOFA without Schmidt orthogonalization (w/o Orth). The corresponding learning curves are presented in Fig. 4. Across all scenarios in Fig. 4, removing sparsemax leads to a noticeable decrease in win rates. The original sparsemax mechanism produces selective and sparse activation patterns, which helps reduce the influence of less relevant features. Without this mechanism, the model exhibits slower improvement and lower final performance, suggesting that sparsity contributes to more stable learning in multi-task settings. A similar trend is observed when Schmidt orthogonalization is removed. The variant without orthogonalization shows consistently lower win rates, indicating that enforcing orthogonality among expert representations helps reduce interference during training. The effect of removing mutual information loss is also reflected in Fig. 4. The w/o MI loss variant displays a clear decline in performance, particularly in scenarios with higher task diversity, where the win rates remain well below those of the complete MOFA model. This suggests that the mutual information constraint supports more consistent optimization across tasks. Overall, the ablation curves highlight that each component contributes to MOFA's performance, and removing any of them leads to reduced learning efficiency or lower final win rates.

Conclusion

This paper proposes a novel multi-task generalization MARL method for non-stationary scenarios, termed MOFA. Built on the CTDE value function decomposition framework, MOFA integrates a MoE module with Schmidt orthogonalization into the agent architecture. This design enforces orthogonal constraints to obtain factorized representations of policy skills. As a result, MOFA decouples generalizable policy skills across tasks and enables zero-shot generalization to unseen tasks through a dynamic policy combination mechanism. We evaluate MOFA in SMAC and PP environments, and the experimental results show that MOFA achieves consistently stronger performance than representative state-of-the-art methods in both multi-task learning and zero-shot generalization settings, demonstrating its adaptability in complex non-stationary environments.

In the future, MOFA's modular design offers opportunities for further extensions. For example, it may be combined with edge computing deployment scenarios, where agents must operate under strict computational or communication constraints, or applied to other distributed multi-agent systems that require efficient knowledge sharing across tasks. These directions provide promising avenues for extending the applicability of MOFA beyond the settings examined in this paper. However, it should be noted that the current experiments were conducted only in simulation environments, and further optimization may be required for real-world scenarios.

Appendix A Parameters of MOFA used in the experiments

The network parameters and training configurations of MOFA are shown in Table 3.

Table 3 Network and training configurations

Configurations	Parameter	Value
Network Configurations	MLP layer dim	64
	Input embedding layer dim	64
	Global encoder dim	64
	Local decoder dim	64
	Hyper-network dim	64
	GRU layer dim	64
	Optimizer	Adam
	Adam α	0.99
	Adam ϵ_R	1×10^{-5}
Training Configurations	Learning rate ϵ	0.0005
	Replay buffer	32
	Batch size	32
	Testing interval	2000
	Target update interval	200
	Maximum timesteps	3,050,000
	Exploration rate	1.0 to 0.05
	Discount factor γ	0.99
	Scaling factor λ	0.01
	Grad norm clip	10
	The number of experts k	3 (SMAC) / 6 (PP)
	α_{MI}	0.001

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Data Availability The data used to support the findings of this study is available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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